

Jan Gebert

Machine Learning Engineer / Data Scientist / Data Engineer / Petroleum Engineer

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SUMMARY

Highly skilled Data Science and Petroleum Engineering Professional with over 5 years of experience, blending expertise in predictive modeling, statistical analysis, and reservoir engineering to address complex challenges in the Energy, E-commerce, and Healthcare sectors.

Proven ability in applying machine learning and statistical methods to enhance petroleum reservoir management, including doubling simulation speeds and slashing analytical times from 2 hours to under 5 minutes.

Successfully implemented a churn prediction model, boosting customer retention by 20% in the Health-Insurance industry. Demonstrated leadership in guiding technical teams through data warehouse architecture projects, fostering innovation and professional development. Excel at deriving actionable insights from complex data to drive strategic decisions.

KEY ASSETS

Reservoir Modelling | Consumer Analytics | Data Science Product Development | Data Pipelines | Advanced Machine Learning | Predictive Modeling | Model Validation | Data Visualization | Leadership | Communication & Teamwork | Problem solving | Creativity & Work Ethic | Coding

PROFESSIONAL EXPERIENCE

Data Scientist

June 2024 – Present

VAPAR Innovation

Brisbane, Australia

Data Scientist in charge of designing, implementing and maintaining Deep Learning models for pipe defect detection from inspection footage. Also in charge of implementing automated gathering of training data from user feedback.

Key Responsibilities:

- Developing AI models for pipe defect detection from inspection footage.
- Building data pipelines to automate the collection and analysis of changes to AI model outputs.
- Performing data analysis to identify performance of AI models against customer expectations.
- Producing insights from customer data to identify opportunities for value creation.
- Build tools to reduce occurrences of errors and improve customer experience.
- Develop supporting algorithms to improve the process of reporting and extracting insights.
- Work closely with web developers, product owners and the business representatives to manage the implementation and delivery of high-quality deep learning models.
- Apply best practice frameworks for model development and lifecycle monitoring.

Tools: Python, Azure, Databricks, VSCode, Jira, Confluence

Career Break

August 2023 – May 2024

Planned career break focused on personal well-being, family commitments, and targeted technical upskilling through hands-on personal projects.

Key Activities & Outcomes:

- Expanded software engineering skill set with C# and Unity, developing a music synthesizer and [fractal visualization](#).
- Strengthened full-stack and web development foundations using JavaScript, including the design and deployment of a [personal website](#).
- Built and deployed multiple end-to-end personal projects, emphasizing data extraction, automation, and visualization ([see GitHub](#)).
- Maintained continuous learning habits through independent research, experimentation, and technical reading.

Data Scientist / Data Engineer

February 2021 – August 2023

Merkur Innovation Lab

Graz, Austria

Data Scientist/Engineer in charge of designing, implementing and maintaining the data warehouse and machine learning models of the corporate start-up. As the team grew from 3 to 11 members, mentored and led the Data Engineering team.

Key Responsibilities:

- Design, implementation and management of data architecture of the data warehouse.
- Implementation, maintenance and testing of ETLs sourcing different Databases and more than 150 tables.
- Ensuring data accuracy and reliability via verifying, cleansing and testing as well as constancy of datasets and metrics.
- Onboarding and mentoring of new data engineering team members, helping them achieve their potential.

Tools: Python, Gitlab CI/CD, Oracle, Microsoft Server SQL, Jira, Confluence

Data Scientist

February 2020 – January 2021

Tiendanube S.A.

Buenos Aires, Argentina

Data Scientist working closely with the Head of Product providing data solutions for a data-driven development of the E-commerce platform.

Key Responsibilities:

- Designed, implemented and automated dashboards for C-Level Management to assist them making data-driven decisions.
- Designed and implemented a bug prioritization system using GitHub API and reporting with Google Sheets and Slack, saving the Head of Product 30 mins daily work.
- Developed an internal python package to identify failed pipelines across the Data Team.

Tools: Python, Postgresql, Git, Asana, AWS, Slack, Microsoft Visual Studio Code

Reservoir Engineer

September 2018 – January 2020

SLB

Buenos Aires, Argentina

Specialized in predictive modeling of petroleum reservoirs, leveraging advanced data visualization techniques to facilitate data-driven decision-making among stakeholders. Demonstrated expertise in building and maintaining strong client relationships through technical support, workshops, and direct client engagement, with a focus on solutions like ECLIPSE, INTERSECT, and Petrel. Proficient in Python and Visual Basic scripting, ensuring the accuracy and reliability of simulation data.

Key Responsibilities:

- **Troubleshooting & Technical Support:** Successfully addressed and resolved complex simulation challenges, including:
 - Led the troubleshooting and support for an offshore horizontal well waterflooding simulation using Petrel and Eclipse in the Gulf of Mexico, overcoming challenges related to simulating pressure drops along the well via advanced well segmentation techniques.
 - Migration of a Polymer flood case from Eclipse to Intersect for an onshore reservoir in San Jorge Gulf, doubling simulation speed.
 - Executed comprehensive history matching for a Waterflooding project in San Jorge Gulf, ensuring accurate simulation of onshore reservoir behavior.
 - Provided specialized support for Shale Fracture Simulation with Kinetix and Intersect in the Vaca Muerta shale reservoir, contributing to optimized fracturing processes.
- **Training & Client Engagement:** Developed and delivered multiple workshops and courses on Petrel RE and Eclipse across Buenos Aires, Mendoza, Neuquén, and Comodoro Rivadavia, strengthening client capabilities and fostering productive relationships.

Tools: Petrel, ECLIPSE, INTERSECT, Kinetix, Python, Visual Basic

Reservoir Engineer Jr.

May 2018 – September 2018

V. y P. Consultores

Buenos Aires, Argentina

Focused on the analytic modeling and data visualization of a heavy oil reservoir in Colombia, employing Sahara and Excel to optimize simulation processes and enhance data management. Achieved a remarkable reduction in analytical simulation time, from 2 hours to under 5 minutes, through efficient VBA scripting.

Key Responsibilities:

- **Analytic Modeling:** Conducted detailed 2D Waterflooding simulations using Sahara to evaluate and optimize water injection strategies for enhancing oil recovery.
- **Data Interpretation:** Skillfully interpreted well logs and production data to derive insights on reservoir characteristics and performance, informing reservoir management decisions.
- **Software Development:** Developed custom VBA applications for streamlined data management and simulations, significantly improving process efficiency and simulation turnaround times.

Tools: Sahara, Visual Basic, Excel

Reservoir Engineer Internship

September 2017 – January 2018

Y.P.F.

Buenos Aires, Argentina

Focused on dynamic reservoir simulation and Enhanced Oil Recovery (E.O.R.) studies, with a primary project involving waterflooding simulation in Mendoza, Argentina. Assessed suitability for polymer injection to improve oil recovery efficiency.

- **History Matching:** Employed advanced simulation techniques to align model predictions with actual field performance, ensuring accurate representation of reservoir behavior for reliable forecasting and strategic planning.
- **Reservoir Monitoring:** Implemented continuous reservoir performance analyses, identifying trends and anomalies to guide operational adjustments and optimize production.
- **Data Preparation and Analysis:** Leveraged Excel Macros for efficient data organization, manipulation, and analysis, enhancing the accuracy and speed of simulations and studies.

Tools: Petrel, Eclipse, T-Navigator, Excel, VBA

Drilling Internship

July 2017 – August 2017

Pan American Energy

Neuquen, Argentina

Acquired comprehensive knowledge of drilling operations in the Neuquen Basin, focusing on both vertical wells in conventional reservoirs and advanced horizontal wells (1500 to 2000 horizontal well length) targeting tight formations.

Developed and implemented VBA scripts to enhance drilling parameter visualization, facilitating real-time decision-making and optimization. Key parameters included Rate of Penetration (ROP), Hook Load, Bit Weight, Flow In Rate, Pump Pressure, Top Drive RPM, and Top Drive Torque.

Tools: Excel, VBA

Reservoir Simulation Research Assistant

May 2017 – July 2017

Reservoir Simulation Center I.T.B.A.

Buenos Aires, Argentina

Proof of Concept (PoC) simulations focused on optimizing lithium extraction from salars in Northern Argentina, a key initiative for sustainable resource management and production.

Utilized advanced simulation software (Eclipse and T-Navigator) to model brine flow dynamics and lithium production processes, enhancing the understanding of subsurface behaviors and extraction potentials.

Tools: Eclipse, T-Navigator

EDUCATION

MSc. in Data and Information Science <i>Fachhochschule Joanneum</i>	2020-2022 <i>Graz - Austria</i>
Diploma in Applied Data Science <i>Universidad Austral</i>	2019 <i>Buenos Aires - Argentina</i>
Petroleum Engineer (MSc. equivalent) <i>Instituto Tecnológico de Buenos Aires</i>	2012-2018 <i>Buenos Aires - Argentina</i>

CERTIFICATIONS

DP-203 Azure Data Engineer Associate	April 2024
BlueEDGE 2 Reservoir Engineering <i>SLB Young Professional Course</i>	June 2019 - July 2019 <i>European Learning Center, Melun - France</i>
Course Content: <i>Petrel Advanced Property Modelling, PVT Fluid Modeling, Eclipse Compositional, Intersect Compositional, MEPO (multiple realization optimizer)</i>	
BlueEDGE 1 Reservoir Engineering <i>SLB Young Professional Course</i>	September 2018 - October 2018 <i>European Learning Center, Melun - France</i>
Course Content: <i>Applied Reservoir Simulation, Eclipse Black Oil, Petrel RE, Petrel RE U&O, Petrel PI, IX (Intersect) lite</i>	
Train The Trainer <i>NExT (a SLB company)</i>	June 2019 <i>Buenos Aires - Argentina</i>
Fracturing Shales <i>NExT (a SLB company)</i>	February 2019 <i>Buenos Aires - Argentina</i>
Gas Well Testing <i>IHRDC</i>	January 2019 <i>e-Learning</i>
Effective Project Team Members <i>MY Consultants Ltd</i>	July 2019 <i>European Learning Center, Melun - France</i>

TECHNICAL SKILLS

Languages: English (fluent), Spanish (bilingual), German (bilingual)
Coding Languages: Python, R, TSQL, SQL, REST API, Postman, Javascript, C#
Databases: PostgreSQL, MySQL, SQLServer, SQLite, Azure Databricks, Oracle, No-SQL
Data Visualization: Matplotlib, Pyplot, Dash, Tableau, PowerBI.
Petroleum Engineering Tools: Petrel, Eclipse, Intersect, Kinetix, Sahara
Misc: Agile methodologies, Jira, Git, Confluence